



Technical Documentation

**ADEC
for series 2000 PLD
and series 4000**

Application: Genset

**Functional Description
Operating Instructions
Workshop Manual
Installation and Initial Operation**

E532233/02E

Printed in Germany

© 2022 Copyright Rolls-Royce Solutions GmbH

Diese Veröffentlichung einschließlich aller ihrer Teile ist urheberrechtlich geschützt. Jede Verwertung oder Nutzung bedarf der vorherigen schriftlichen Zustimmung der Rolls-Royce Solutions GmbH. Das gilt insbesondere für Vervielfältigung, Verbreitung, Bearbeitung, Übersetzung, Mikroverfilmungen und die Einspeicherung und / oder Verarbeitung in elektronischen Systemen, einschließlich Datenbanken und Online-Diensten. Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen. Änderungen bleiben vorbehalten.

Printed in Germany

© 2022 Copyright Rolls-Royce Solutions GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of Rolls-Royce Solutions GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Imprimé en Allemagne

© 2022 Copyright Rolls-Royce Solutions GmbH

Tout droit réservé pour cet ouvrage dans son intégralité. Toute utilisation ou exploitation requiert au préalable l'accord écrit de Rolls-Royce Solutions GmbH. Ceci s'applique notamment à la reproduction, la diffusion, la modification, la traduction, l'archivage sur microfiches, la mémorisation et / ou le traitement sur des systèmes électroniques, y compris les bases de données et les services en ligne.

Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

Impreso en Alemania

© 2022 Copyright Rolls-Royce Solutions GmbH

Esta publicación se encuentra protegida, en toda su extensión, por los derechos de autor. Cualquier utilización de la misma, así como su reproducción, difusión, transformación, traducción, microfilmación, grabación y/o procesamiento en sistemas electrónicos, entre los que se incluyen bancos de datos y servicios en línea, precisa de la autorización previa de Rolls-Royce Solutions GmbH.

El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

Nos reservamos el derecho de introducir modificaciones.

Stampato in Germania

© 2022 Copyright Rolls-Royce Solutions GmbH

Questa pubblicazione è protetta dal diritto d'autore in tutte le sue parti. Ciascun impiego o utilizzo, con particolare riguardo alla riproduzione, alla diffusione, alla modifica, alla traduzione, all'archiviazione in microfilm e alla memorizzazione o all'elaborazione in sistemi elettronici, comprese banche dati e servizi on line, deve essere espressamente autorizzato per iscritto dalla Rolls-Royce Solutions GmbH.

Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

Con riserva di modifiche.

Impresso na Alemanha

© 2022 Copyright Rolls-Royce Solutions GmbH

A presente publicação, inclusive todas as suas partes, está protegida pelo direito autoral. Qualquer aproveitamento ou uso exige a autorização prévia e por escrito da Rolls-Royce Solutions GmbH. Isto diz respeito em particular à reprodução, divulgação, tratamento, tradução, microfilmagem, e a memorização e/ou processamento em sistemas eletrônicos, inclusive bancos de dados e serviços on-line.

Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

Functional Description	03
1 Use, Structure and Function	05
1.1 Use	05
1.2 ECS-5 – Use of devices	13
1.3 ADEC – Functions	14
1.4 Display DIS 10 – Structure	29
2 Technical Data	33
2.1 SAM – Technical data	33
2.2 Engine governor	37
2.3 Analog display instruments	39
2.4 CCB 2 – Technical Data	40
Operating Instructions	41
3 Troubleshooting	43
3.1 Fault indication on SAM display – Genset applications	43
4 View of Windows Web Page	67
4.1 Engine governor web feature	67
5 Tasks	69
5.1 Engine wiring check	69
5.2 ECU – Self-test implementation	70
5.3 SAM – Self-test	71
Workshop Manual	73
6 Repair Work	75
6.1 SAM – Replacement	75
6.2 SAM fuse – Replacement	76
Installation and Initial Operation	77
7 Mechanical Installation	79
7.1 Routing cables between the installation locations	79
7.2 SAM – Installation	80
7.3 DIS – Installation	81
8 Electrical Installation	83
8.1 Electrical installation	83
8.2 Engine governor signals	92
8.3 SAM signals	111
8.4 Engine governor - Connection	178

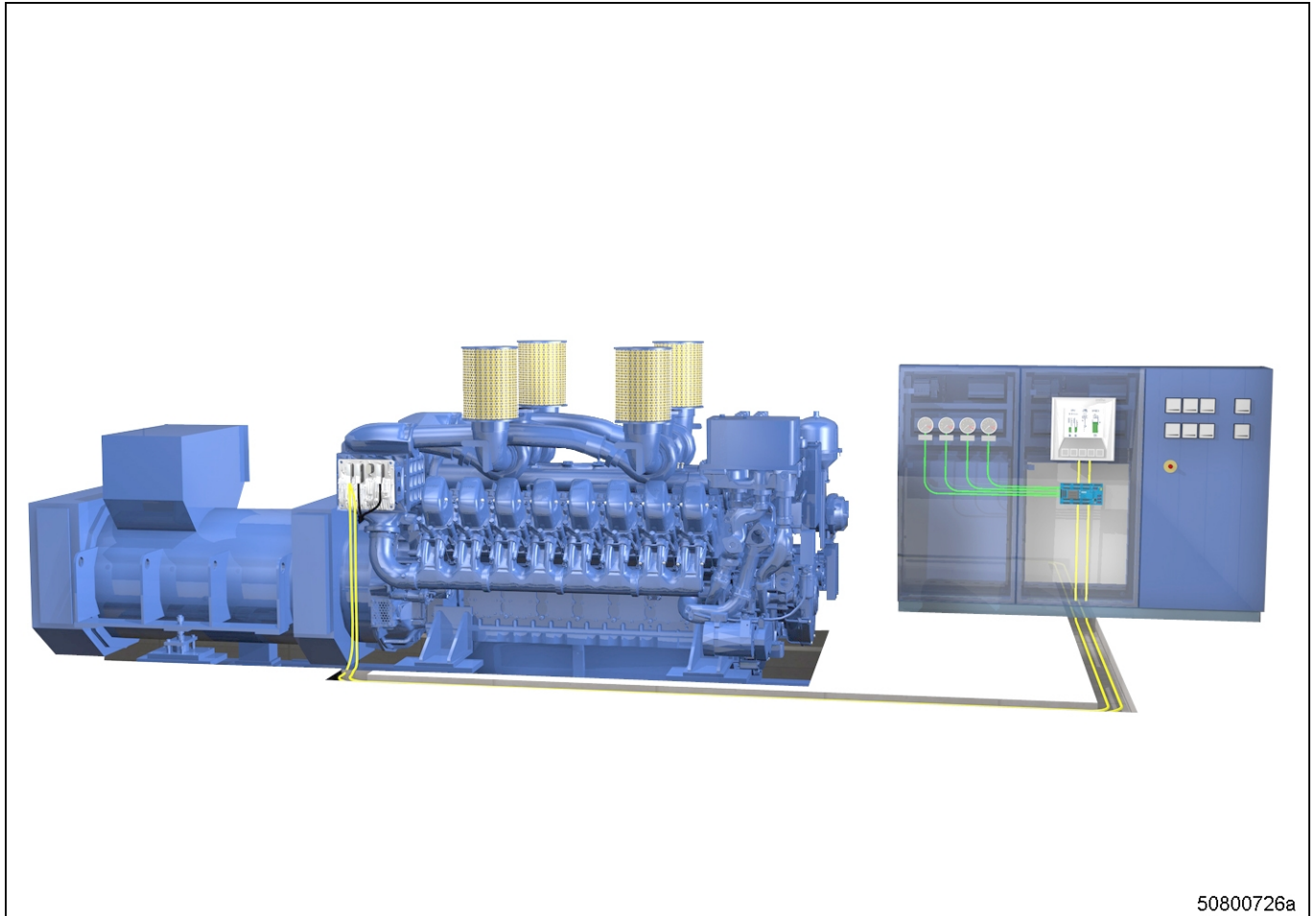
9	Initial Operation	179
9.1	Checks prior to startup	179
9.2	Operating voltage – Initial application	180
9.3	SAM power supply – Check	181
10	Settings	183
10.1	CAN node – Configuration	183
10.2	Parameters – Setting with dialog unit	184
10.3	SAM MINIDIALOG settings	186
10.4	Additional SAM parameters and options	188
10.5	Index	193

Functional Description

1 Use, Structure and Function

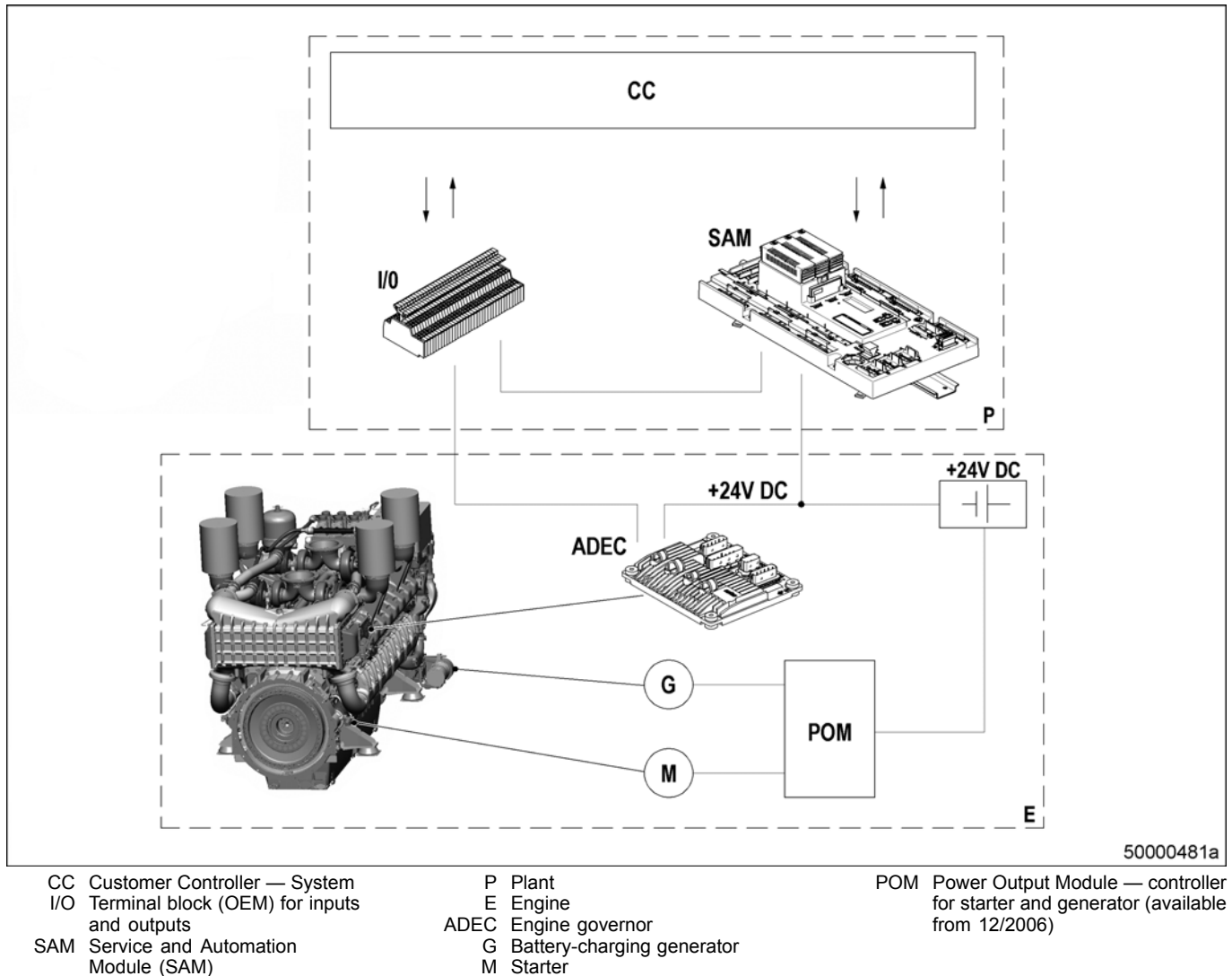
1.1 Use

Functions



50800726a

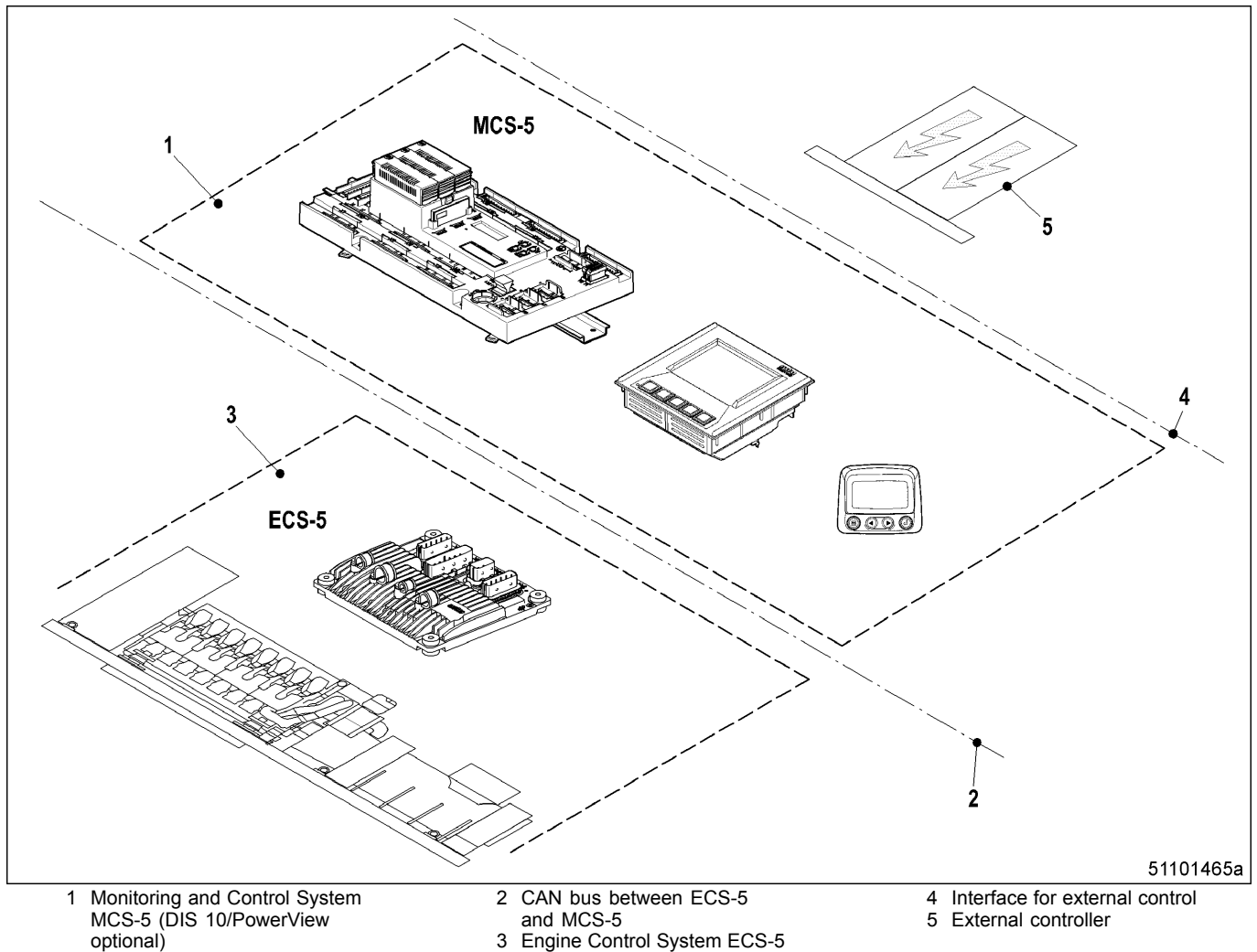
- Control of the engine.
- Monitoring of operating states.
- Closed-loop control of fuel injection and engine speed (depending on operating state).
- Indication of faulty operating states (display SAM).



Features

- Electronic engine governing and control.
- Monitoring of inadmissible engine operating states.
- Display of fault messages and fault codes.
- Connecting cable for power supply to engine governor.
- Connecting cable for connection to a higher-level genset control system.
- Hardware interfaces to a higher-level control system (option).
- Inputs for plant sensors (option).
- Engine safety features including engine shutdown.
- Integral fault diagnosis system ITS.
- Integral load profile recorder.
- Speed droop switching possible with engine running.
- Straightforward engine governor replacement.
 - Engine and interface data stored in SAM.
 - Complete automatic software download following connection of a new, unprogrammed engine governor.
- ECMS — Engine Side Condition Management System
 - Automatic engine power reduction as a function of:
 - Intake air temperature
 - Operating site (e.g. altitude above sea level)
 - Other parameters

Design of the overall system



The design of the overall system comprising the ECS-5 and MCS-5 subsystems depends on customer requirements and the higher-level control system.

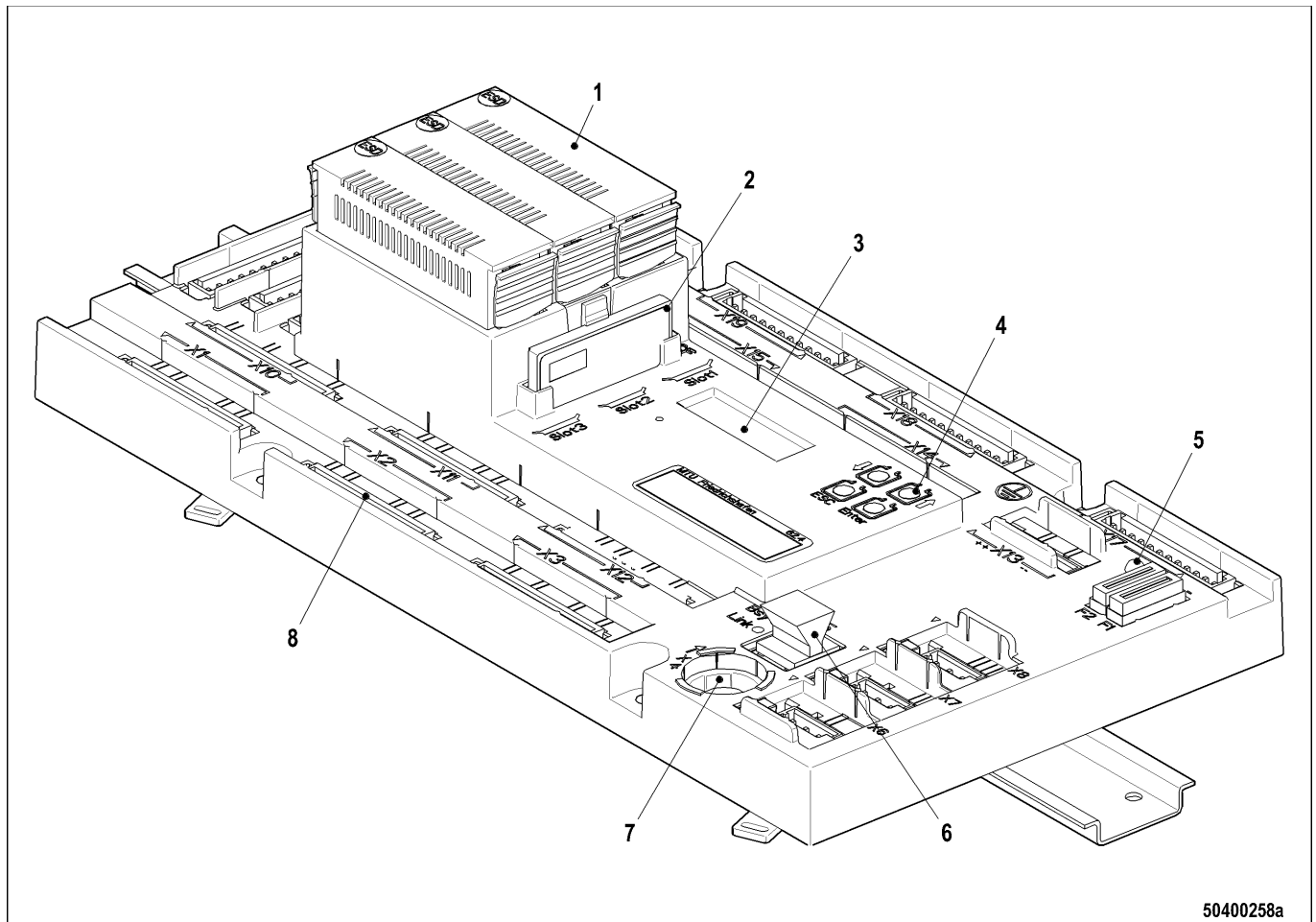
Design of Engine Control System ECS-5

The Engine Control System comprises the following devices:

- Engine governor
- Sensors on engine
- Actuators on engine
- Injectors on engine
- Wiring harnesses on engine

Basic scope of Monitoring and Control System MCS-5

Service and Automation Module (SAM)



- 1 Module cassette, slots for additional I/O PIM cards (CCB2 for CANOpen and J1939, optional)
2 Compact flash memory card

- 3 Display for fault codes and minialog
4 Control keys for minialog
5 Diagnostic lamp

- 6 Ethernet (with protective cap)
7 Interface for dialog unit
8 Connector with spring design

Functions

SAM functions

- Display of fault codes from engine governor and SAM (3).
- Backup function, engine life data are stored
 - every hour,
 - after every engine stop
 - after every emergency engine stop.
- Interface for dialog unit.

Diagnosis

- Straightforward diagnosis by fault code display.
- Self-diagnosis by diagnostic lamp (5).
 - Steady = SAM is OK.
 - Flashing = SAM is faulty, contact Service. The flashing codes mean:
 -
 - Dark = Supply voltage missing.

Customer interface

- 24 binary outputs
- 3 PWM outputs
- 8 display outputs